

# Cumulative Test-1

Physics(Set - W)

Date: September 29, 2026

Full Marks: 40

Class – XI

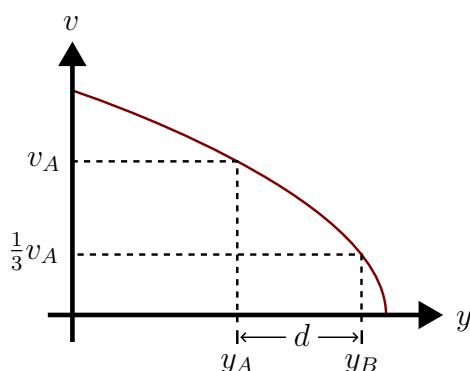
Time: 20.0 Minutes

## General Instructions

1. **Negative Marking** is  $-1$  for a single answer wrong
2. Use **blue** or **black** ink only.
3. Mobile phones, calculators, and other electronic devices are not permitted.
4. There is only Section A but in your OMR fill the questions 21 to 25 under Section B.

## SECTION A: Single Correct Answer Type

1. Figure shows the speed  $v$  versus height  $y$  of a ball tossed directly upward, along a  $y$  axis. Distance  $d$  is  $0.40$  m. The speed at height  $y_A$  is  $v_A$ . The speed at height  $y_B$  is  $\frac{1}{3}v_A$ . What is speed  $v_A$ ?



- (A)  $2.6 \text{ m s}^{-1}$       (B)  $2.7 \text{ m s}^{-1}$   
(C)  $3.0 \text{ m s}^{-1}$       (D)  $8.9 \text{ m s}^{-1}$
2. A man releases a stone at the top edge of a tower. During the last second of its travel,

the stone falls through a distance of  $\frac{9}{25}H$ , where  $H$  is the tower's height. Find  $H$ .

- (A) 100 m      (B) 122.5 m  
(C) 145 m      (D) 167.5 m

3. A key falls from a bridge that is  $45$  m above the water. It falls directly into a model boat, moving with constant velocity, that is  $12$  m from the point of impact when the key is released. What is the speed of the boat?

- (A)  $2.6 \text{ m s}^{-1}$       (B)  $3.0 \text{ m s}^{-1}$   
(C)  $4.0 \text{ m s}^{-1}$       (D)  $5.0 \text{ m s}^{-1}$

4. A hoodlum throws a stone vertically downward with an initial speed of  $15.0 \text{ m s}^{-1}$  from the roof of a building,  $30.0$  m above the ground. How long does it take the stone to reach the ground, and what is the speed of the stone at impact?

- (A) 1.38 s,  $28.5 \text{ m s}^{-1}$
- (B) 1.38 s,  $15.0 \text{ m s}^{-1}$
- (C) 2.45 s,  $39.0 \text{ m s}^{-1}$
- (D) 1.75 s,  $32.2 \text{ m s}^{-1}$

5. A melon is dropped from a height of 39.2 m. After it crosses through half that distance, the acceleration due to gravity is reduced to zero by air drag. With what velocity does the melon hit the ground?

- (A)  $9.8 \text{ m s}^{-1}$                       (B)  $14.0 \text{ m s}^{-1}$
- (C)  $19.6 \text{ m s}^{-1}$                       (D)  $28.0 \text{ m s}^{-1}$

6. A world's land speed record was set by Colonel John P. Stapp when in March 1954 he rode a rocket-propelled sled that moved along a track at  $1020 \text{ km h}^{-1}$ .



He and the sled were brought to a stop in 1.4 s. In terms of  $g$ , what acceleration did he experience while stopping?

- (A)  $10g$                                       (B)  $14g$
- (C)  $21g$                                       (D)  $29g$

7. In a particle accelerator, an electron enters a region in which it accelerates uniformly in a straight line from a speed of  $4.00 \times 10^5 \text{ m s}^{-1}$  to a speed of  $6.00 \times 10^7 \text{ m s}^{-1}$  in a distance of 3.00 cm. For what time interval does the electron accelerate?

- (A)  $4.97 \times 10^{-10} \text{ s}$     (B)  $9.93 \times 10^{-10} \text{ s}$
- (C)  $1.49 \times 10^{-9} \text{ s}$     (D)  $2.98 \times 10^{-9} \text{ s}$

8. You are arguing over a cell phone while trailing an unmarked police car by 25 m; both your car and the police car are traveling at  $120 \text{ km h}^{-1}$ . Your argument diverts your attention from the police car for 2.0 s (long enough for you to look at the phone and yell, "I won't do that!"). At the beginning of that 2.0 s, the police officer begins braking suddenly at  $5.0 \text{ m s}^{-2}$ . (a) What is the separation between the two cars when your attention finally returns? Suppose that you take another 0.40 s to realize your danger and begin braking. (b) If you too brake at  $5.0 \text{ m s}^{-2}$ , what is your speed when you hit the police car?

- (A) 15.0 m,  $28.9 \text{ m s}^{-1}$
- (B) 15.0 m,  $33.3 \text{ m s}^{-1}$
- (C) 10.6 m,  $28.9 \text{ m s}^{-1}$
- (D) 10.0 m,  $25.0 \text{ m s}^{-1}$

9. A rocket, initially at rest, is fired vertically with an upward acceleration of  $10.0 \text{ m s}^{-2}$ . At an altitude of 0.500 km, the engine of the rocket cuts off. What is the maximum altitude it achieves?

- (A) 1.9 km                                      (B) 1.3 km
- (C) 1.6 km                                      (D) 1.0 km

10. A stone is thrown from the top of a building with an initial velocity of  $20 \text{ m s}^{-1}$  downward. The top of the building is 60 m above the ground. How much time elapses between the instant of release and the instant of impact with the ground?

- (A) 2.0 s                                              (B) 6.1 s
- (C) 3.5 s                                              (D) 1.6 s

## Answer Key

- |        |        |        |         |
|--------|--------|--------|---------|
| 1. (C) | 4. (A) | 7. (B) | 10. (A) |
| 2. (B) | 5. (C) | 8. (A) |         |
| 3. (C) | 6. (C) | 9. (D) |         |